

GenVarX Suite

Genomic Variant Effect Predictor & Clinical Analysis Platform

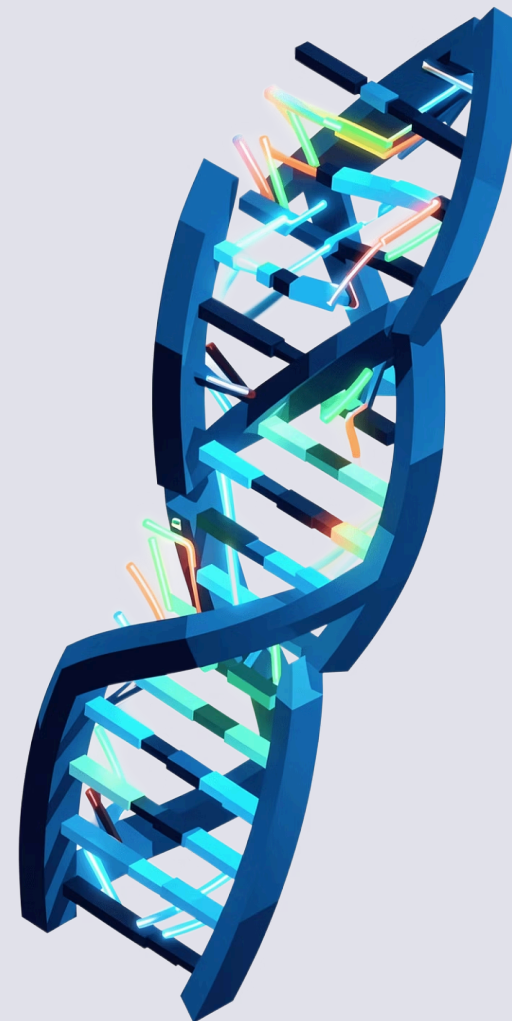
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The Problem: Interpretation Gap

Time-Consuming & Error-Prone

Clinical variant interpretation demands extensive manual effort, introducing inconsistency and risk at every step.

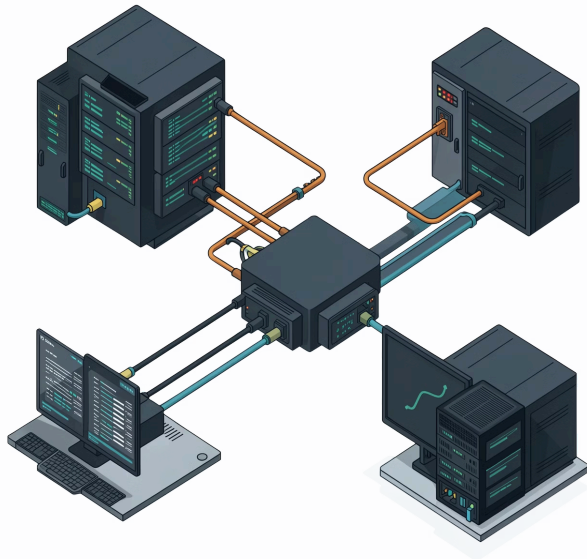
Siloed Databases

Researchers must manually cross-reference GenBank, EMBL, and SwissProt – with no shared context between systems.

No Unified Resolution

No existing system connects genomic coordinates directly to clinical significance in a single, automated workflow.



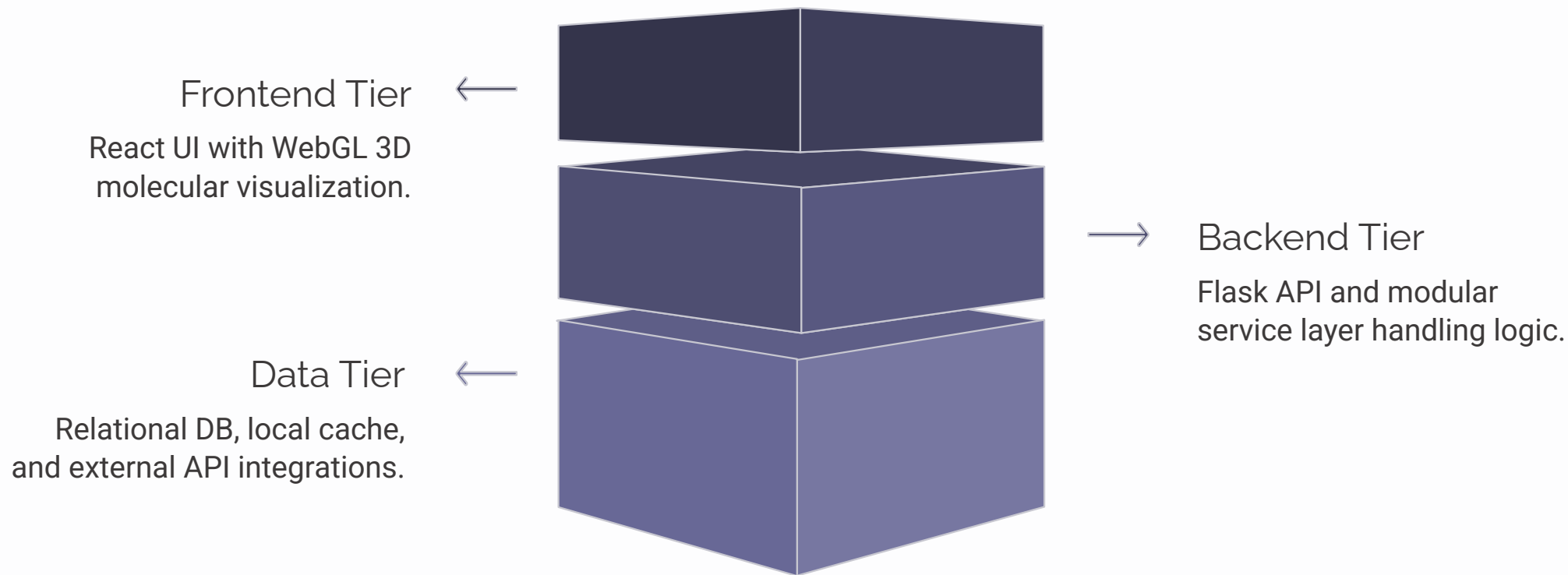


The Solution: Automated Pipeline

GenVarX Suite delivers an end-to-end variant interpretation workflow — from raw genomic data through to clinical decision-making — without manual hand-offs.

- Integrates sequence alignment, machine learning, and medical databases in one unified system
- Bridges **ClinVar**, **GWAS Catalog**, and **ChEMBL** for comprehensive multi-source analysis
- Reduces researcher burden by automating cross-database resolution and annotation

System Architecture



A clean separation of concerns across three layers ensures modularity, scalability, and maintainability – each tier communicating through well-defined interfaces.

Core Processing Pipeline

1

Input

Genomic coordinates or Reference SNP IDs (RSIDs)

2

Alignment

Needleman-Wunsch or Smith-Waterman algorithms

3

AI Classification

SIFT/PolyPhen predictive modeling for mutation scoring

4

Clinical Annotation

ClinVar and GWAS database enrichment

Technology Stack



Languages

Python & Perl power backend automation, scripting, and data processing workflows.



Visualization

WebGL2 renders interactive 3D molecular structures and mutation impact assessments.



Machine Learning

Hidden Markov Models and neural networks drive variant effect prediction at scale.



Database

Relational model with **ASN.1** data exchange format for interoperability with NCBI and EBI.

Key Features

Variant Identification

Automated coordinate-to-RSID resolution eliminates manual lookup.

Predictive Classification

Mutations scored and labeled as **pathogenic** or **benign** via ML models.

Pharmacogenomics

Integrated drug discovery pipeline for targeted therapeutic matching.

Real-Time Querying

Live access to NCBI and EBI repositories keeps annotations current.



THANK YOU

